SQL

- Learn the Basics
  - What are Relational Databases?
  - RDBMS Benefits and Limitations
  - SQL vs NoSQL Databases
- Basic SQL Syntax
  - SQL Keywords
  - Data Types
  - Operators
  - Statements: SELECT, INSERT, DELETE, UPDATE
- Data Definition Language (DDL)
  - Truncate Table
  - Alter Table
  - Create Table
  - Drop Table
- Data Manipulation Language (DML)
  - Select
  - FROM, WHERE, JOINs, GROUP BY, ORDER BY, HAVING, INSERT, UPDATE, DELETE
- Aggregate Queries
  - SUM, COUNT, AVG, MIN, MAX, GROUP BY, HAVING
- Data Constraints
  - Primary Key
  - Foreign Key
  - Unique
  - NOT NULL
  - CHECK
- Different Types
  - Scalar, Column, Row, Table
- Subqueries
  - Nested Subqueries
  - Correlated Subqueries
- JOIN Queries
  - INNER JOIN
  - LEFT JOIN
  - RIGHT JOIN
  - FULL OUTER JOIN
  - Self Join
  - Cross Join
- Advanced Functions
  - String Functions: CONCAT, LENGTH, SUBSTRING, REPLACE, UPPER, LOWER
  - Date and Time: DATE, TIME, TIMESTAMP, DATEPART, DATEADD
  - Numeric Functions: FLOOR, ABS, MOD, ROUND, CEILING
  - Conditional: CASE, NULLIF, COALESCE
- Views
  - Creating Views
  - Modifying Views
  - Dropping Views
- Managing Indexes
- Query Optimization
- Indexes
- Data Integrity Constraints
  - GRANT and Revoke
  - DB Security Best Practices
- Transaction Isolation Levels
- Transactions
  - ACID
  - BEGIN
  - COMMIT
  - ROLLBACK
  - SAVEPOINT
- Stored Procedures & Functions
- Data Integrity & Security
- Performance Optimization
  - Query Analysis Techniques
  - Query Optimization Techniques: Using Indexes, Optimizing Joins, Reducing Subqueries, Selective Projection
- Advanced SQL
  - rank
  - dense_rank
  - lead, lag
  - Row_number
  - Window Functions
  - Recursive Queries
  - Pivot / Unpivot Operations
  - Common Table Expressions
  - Dynamic SQL

Have a look at the following related roadmaps

- Backend
- PostgreSQL
- MongoDB